Original Article

A Hospital-Based Study on the Distribution of Ovarian Tumors in the Tribal and Nontribal Population of Ranchi, Jharkhand

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Abstract

Objectives: This study aimed to determine the types and management of ovarian tumors in tribal and nontribal population of Jharkhand. **Methods:** This prospective study was conducted in the Department of Pathology for duration of 24 months from December 2019 till November 2021. The study included the data (age and clinical symptoms) of patients of ovarian tumors who attended the Gynecology Outpatient Department and were managed with surgery or neoadjuvant therapy. The diagnosis was confirmed through radiology investigations, tumor markers, and histopathology. Histopathological reports of samples sent to the Department of Pathology were included. **Results:** A total of 150 patients with ovarian tumors were included, among which 63 (42%) were tribal. As compared to nontribals, tribals had comparable mean age (39.05 ± 12.18 vs. 36.55 ± 11.77, P = 0.469), comparable parity, and socioeconomic status. The most common complaints were pain in lower abdomen (80%), lump in lower abdomen (34%), and abdominal distension (18%). Epithelial tumors constituted the majority of tumors, and histopathological types included mucinous adenocarcinoma (26%), dermoid cyst (16%), serous cystadenoma (10%), and simple ovarian cyst (10%). Distribution of histopathological type of tumor was comparable between nontribal and tribal (P = 0.478). Seventy-four percent of the patients were managed operatively, which included unilateral ovarian mass/cyst removal, ovarian mass removal, total abdominal hysterectomy, and opposite side salpingo-oophorectomy, whereas 26% of the patients were managed conservatively with chemotherapy. There was no mortality, and the overall outcomes were comparable between nontribals and tribals (P > 0.05). **Conclusion:** The demographic characteristics, clinical presentations, histopathological type, management, and outcomes were similar among tribal and nontribal patients with ovarian tumors.

Keywords: *Mucinous adenocarcinoma, ovarian tumor, oophorectomy, tribal*

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Introduction

Ovarian tumors are accountable for 6% of all the carcinomas worldwide among females and rank fifth in all tumors. The ovarian tumors rank at third place in female genital tract tumors after cervical and endometrial carcinomas.[1] According to GLOBOCAN 2020, the number of new cases of ovarian tumors was 313,959, which constituted 1.6% of all cancer cases, among which 207,252 patients died, which constituted 2.1% of all deaths due to ovarian cancer.[2]

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It is difficult to diagnose the ovarian tumors at an early stage. Suspicion plays a big role in early diagnosis. As a result, further work is needed to highlight the issues with diagnosis of ovarian tumors at early stage.[1]

Jharkhand is a largely tribal state, with 28% of the population being tribal. The definition of tribal is “a

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group of people or community related to each other with similar ancestors, customs and traditions." Jharkhand is one of the five Indian states with the lowest Human Development Index and has a total of 32 different tribes inhabiting the state.[3,4]

There is a lack of awareness in tribal women regarding various diseases. In addition, there is a considerable paucity of official data on ovarian cancers, especially for tribal women, which is crucial for public health-care planning.

The present study was thus conducted to determine the types and management of ovarian tumors in tribal and nontribal population of Jharkhand presenting to our hospital.

Methods

This prospective study was conducted at the Department of Pathology of Rajendra Institute of Medical Sciences (RIMS), Jharkhand, for duration of 24 months from December 2019 till November 2021 after taking institutional ethical clearance. RIMS is a tertiary care referral hospital in the state of Jharkhand which caters to the whole population of the state. It is an 800-bedded free government hospital of Jharkhand where around 50% of the visiting population is tribal.

Inclusion criteria

The study included patients with ovarian malignancy in whom exploratory laparotomy was done, and those patients in whom surgery was done for either ovarian mass alone or in addition to hysterectomy.

Exclusion criteria

The study excluded patients with pelvic lump arising from uterus or adnexal lump of other etiologies, such as inflammatory lump or chocolate cyst of ovary, and those who were treated conservatively for ovarian tumors.

Taking the study of Deshmukh and Suboohi[1] as reference, the total sample size taken for the study was 150. Written informed consent was obtained from all the study patients.

The diagnosis was confirmed through ultrasound (US), contrast-enhanced computed tomography, tumor markers, US-guided needle aspiration, and frozen section. The operated gross specimen was sent to the Department of Pathology for final histopathological diagnosis. Either staging laparotomy surgery with total abdominal hysterectomy and bilateral salpingectomy was done or neoadjuvant chemotherapy was given in cases of advanced ovarian carcinoma. Histopathological reports of ovarian tumor samples sent to the Department of Pathology were included.

Statistical analysis

The presentation of the categorical variables was done in the form of number and percentage (%). On the other hand, the quantitative data were presented as the means ± standard deviation and as median with 25th and 75th percentiles (interquartile range). The comparison of the variables which were quantitative and not normally distributed in nature was analyzed using Mann–Whitney *U*-test (for two groups), and independent *t*-test was used for comparison of normally distributed data between two groups. The comparison of the variables which were qualitative in nature was analyzed using Chi-square test. If any cell had an expected value of < 5, then Fisher's exact test was used.

The data entry was done in the Microsoft Excel spreadsheet, and the final analysis was done with the use of the Statistical Package for the Social Sciences (SPSS) software, IBM manufacturer, Chicago, Illinois, USA, version 21.0.

Results

A total of 150 patients with ovarian tumors were included, among which 63 (42%) were tribal. As compared to nontribals, tribals had comparable mean age (39.05 ± 12.18 vs. 36.55 ± 11.77, $P = 0.469$); comparable parity, with majority being multiparous (76.19% vs. 58.62%, $P = 0.477$); and comparable socioeconomic status (SES), with majority from low SES (85.71% vs. 79.31%, $P = 0.609$) [Table 1].

The most common complaints were pain in lower abdomen (80%), lump in lower abdomen (34%), and abdominal distension (18%). Other complaints were loss of appetite, burning micturition for 1 day, vomiting, heavy menstrual bleeding, primary infertility with right ovarian mass, weakness in both limbs, retention of urine, watery discharge per vaginum, difficulty in breathing, bloating, irregular menses, amenorrhea, and backache in only 1 out of 50 patients (2%) each. The symptoms were comparable among both the ethnic groups.

Distribution of type of tumor was comparable between nontribal and tribal (benign: 48.28% vs. 61.90%, respectively, and malignant: 51.72% vs. 38.10%, respectively) ($P = 0.34$).

Distribution of histological type was comparable between nontribal and tribal (epithelial tumor: 62.07% vs. 42.86%, respectively; functional ovarian cyst: 13.79% vs. 33.33%, respectively; and germ cell tumor: 24.14% vs. 23.81%, respectively) ($P = 0.237$).

Compared to nontribals, tribal population had lesser incidence of mucinous adenocarcinoma (14.29% vs. 34.48%), serous adenocarcinoma (4.76% vs. 6.90%),

Table 1: Demographic characteristics of the study population

Parameters	Nontribal (*n*=87), *n* (%)	Tribal (*n*=63), *n* (%)	Total, *n* (%)	*P*
Age (years), mean±SD	36.55±11.77	39.05±12.18	37.6±11.9	0.469*
Parity				
Nulli	27 (31.03)	12 (19.05)	39 (26)	0.477‡
Primi	9 (10.34)	3 (4.76)	12 (8)	
Multi	51 (58.62)	48 (76.19)	99 (66)	
Socioeconomic status				
Upper middle	6 (6.90)	0 (0)	6 (4)	0.609‡
Lower middle	12 (13.79)	9 (14.29)	21 (14)	
Low	69 (79.31)	54 (85.71)	123 (82)	

Test name: *Independent *t*-test, ‡Fisher's exact test. SD: Standard deviation

endometrioid carcinoma (0% vs. 3.45%), dermoid cyst (14.29% vs. 17.24%), serous cystadenoma (9.52% vs. 10.34%), simple ovarian cyst (9.52% vs. 10.34%), mucinous cystadenoma (4.76% vs. 6.90%), and endodermal sinus tumor (0% vs. 3.45%) and higher incidence of mucinous cystadenocarcinoma (4.76% vs. 0%), lutein cyst (14.29% vs. 0%), follicular cyst (9.52% vs. 3.45%), dysgerminoma (4.76% vs. 3.45%), immature teratoma (4.76% vs. 0%), and solid clear cell carcinoma (4.76% vs. 0%); however, statistically, there was no significant difference (P = 0.478) [Table 2].

Management

One hundred and one (74%) patients were managed operatively, whereas 39 patients (26%) were managed conservatively with chemotherapy. Surgical management included unilateral ovarian mass/cyst removal with ovarian mass removal (30%), total abdominal hysterectomy and opposite side salpingo-oophorectomy (28%), and staging laparotomy including omentectomy and peritoneal fluid being sent for histopathology (12%) and bilateral ovarian mass/cyst removal in 6 patients (4%).

Outcomes

Distribution of outcome was comparable between nontribal and tribal (patient discharged: 48.28% vs. 61.90%, respectively; patient transferred to the Oncology Department: 13.79% vs. 9.52%, respectively; patient transferred to Oncosurgery Department: 17.24% vs. 19.05%, respectively; and patient discharged with referral to oncology: 20.69% vs. 9.52%, respectively) (P = 0.734).

Discussion

The present study holds importance among the various studies conducted on ovarian tumors since it portrays the characteristics pertaining to tribal and nontribal population. Overall, we found no significant differences in the demographic, clinical characteristics and outcomes of the patients.

Table 2: Tumor type

Histopathology type	Nontribal, *n* (%)	Tribal, *n* (%)	Total, *n* (%)	*P*
Dermoid cyst	15 (17.24)	9 (14.29)	24 (16)	0.478‡
Dysgerminoma	3 (3.45)	3 (4.76)	6 (4)	
Endodermal sinus tumor	3 (3.45)	0	3 (2)	
Endometrioid carcinoma	3 (3.45)	0	3 (2)	
Follicular cyst	3 (3.45)	6 (9.52)	9 (6)	
Immature teratoma	0	3 (4.76)	3 (2)	
Lutein cyst	0	9 (14.29)	9 (6)	
Mucinous adenocarcinoma	30 (34.48)	9 (14.29)	39 (26)	
Mucinous cystadenocarcinoma	0	3 (4.76)	3 (2)	
Mucinous cystadenoma	6 (6.90)	3 (4.76)	9 (6)	
Serous adenocarcinoma	6 (6.90)	3 (4.76)	9 (6)	
Serous cystadenoma	9 (10.34)	6 (9.52)	15 (10)	
Simple ovarian cyst	9 (10.34)	6 (9.52)	15 (10)	
Solid clear cell carcinoma	0	3 (4.76)	3 (2)	
Total	87 (100)	63 (100)	150 (100)	

‡Fisher's exact test

In the present study, the mean age of the participants was 37.6 ± 11.9 years. There was no difference in the mean age of tribals and nontribals (36.55 ± 11.77 vs. 39.05 ± 12.18, P = 0.469). Similarly, in a study including tribal population living in Mumbai, Jha *et al*.[5] reported that the mean age was 32.53 years. Concurrently, Kumar *et al*.,[6] who included tribal patients of Jharkhand, reported that most of the patients (66%) were in the age group of 21–40 years. Age is strongly correlated with risk of ovarian carcinoma, where majority of the ovarian cancer cases (80%) are diagnosed after the age of 50 years, whereas benign cases are diagnosed in the forties. Age carries importance since it is directly correlates with the risk of malignant transformation.[7-9]

There is a significant role of parity in terms of ovarian carcinoma. The differences in cancer incidence between parous and nulliparous women are explained by the cell differentiation, proliferation, and apoptosis.[10,11]

In the present study, tribals had comparable parity with that of nontribals, with majority being multiparous (58.62% vs. 76.19%, $P = 0.477$) and others being nullipara or primi. In line with our study, Deshmukh and Suboohi[1] reported that 46% of the women were para 3–4, followed by 44% being para 1–2, with only 2 women being nullipara. In a study including tribal population of Jharkhand, Kumar *et al*.[6] reported that majority (40%) were para 1–2, followed by 30% women being nullipara (30%). Specific studies have shown.

An association is reported between lower SES and greater risk of most types of malignancy. The association can be explained due to the presence of lifestyle factors, such as cigarette smoking, more physical activity, and higher intake of nonstarchy vegetables.[12-15]

In the present study, no significant difference was found between the SES of tribals and nontribals, Overall, majority of the women were from low SES (79.31% and 85.71%) reinforcing the fact that lower SES increases the risk of malignancy. Further, Præstegaard *et al*.[14] found that there was an association between lower education level and increased risk of advanced tumor stage at the time of diagnosis of ovarian carcinoma. The findings of the study demonstrated socioeconomic differences in stage at diagnosis of ovarian carcinoma.

However, Alberg *et al*.[13] found that risk of ovarian carcinoma was inversely associated with SES in African-American women. Patients who had family annual incomes of ≥$75,000 demonstrated an odds ratio of 0.74 (95% confidence interval: 0.47, 1.16) in comparison to women who had incomes <$10,000 ($P = 0.055$). This may be because of different ethnicity which makes the results of the present study important.

At the time of disease presentation, the majority of the women are symptomatic. The chief symptoms are ascites, bowel obstruction, diarrhea, constipation, nausea, vomiting, and gastrointestinal reflux. Other chief complains are fatigue, shortness of breath, abdominal bloating, abdominal pain, and pelvic pain.[15]

Most of the women in the present study had pain in lower abdomen (80%), lump in lower abdomen (34%), and abdominal distension (18%). In the study by Kumar *et al*.,[6] abdominal mass was present in 60%, vaginal bleeding in 30%, and ascites in 10% of the cases.

Deshmukh and Suboohi[1] found that pain in abdomen was present in 38%, lump in abdomen in 32%, and nausea and vomiting in 14% of the women. Agrawal *et al*.[16] also found that pain in abdomen (50.9%) was present in most of the women, followed by lump in abdomen (29.2%). In women with borderline and malignant tumors, the common complaints were ascites, anorexia, and weight loss.

For the diagnosis, management, and prognosis of ovarian carcinoma, it is important to determine histology pattern. The most common ovarian tumors found in the present study were mucinous adenocarcinoma (26%), dermoid cyst (16%), serous cystadenoma (10%), and simple ovarian cyst (10%). Distribution of histopathological examination (HPE) report was similar between nontribal and tribal. In the study by Deshmukh and Suboohi,[1] out of 70 cases of ovarian cancer, there were 26 cases of serous cystadenocarcinoma, 10 cases of germ cell tumor (immature), and 6 cases of granulosa cell tumor. The borderline malignancy was present in 14 cases. Others tumors were metastatic adenocarcinoma, tiny focus of malignancy found, and Krukenberg tumor. Agrawal *et al*.[16] reported that the surface epithelial tumors were the most common ovarian tumor (72.1%), followed by germ cell tumors (19.9%), sex cord stromal tumors (7.1%), and metastatic (0.2%).

In the present study, there was no significant difference in outcomes among nontribals and tribals, as most of the patients were discharged (48.28% vs. 61.90%), and rest of the patients were transferred to the Departments of Oncology and Oncosurgery. This shows that all patients were managed adequately irrespective of the tribal ethnicity, and all the patients responded similarly irrespective of the tribal residence.

Limitations of the study

The study is limited by single-center data. Second, the mortality and the change in the quality of life of the patients were not assessed in the follow-up period. Third, causative association risk analysis for the tribal population was not done.

Conclusion

Most of the tribal women were in thirties, multipara, and belonged to low SES. Pain in lower abdomen was the chief complaint. Benign tumors were present in most of the cases. Epithelial tumors constituted the majority of tumors. Mucinous adenocarcinoma, dermoid cyst, serous cystadenoma, and simple ovarian cyst were the common histopathological types. Overall, statistically, the demographic characteristics, clinical presentations, histopathological type, management, and outcomes were similar among tribal and nontribal patients with ovarian tumors.

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Conflicts of interest

There are no conflicts of interest.

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